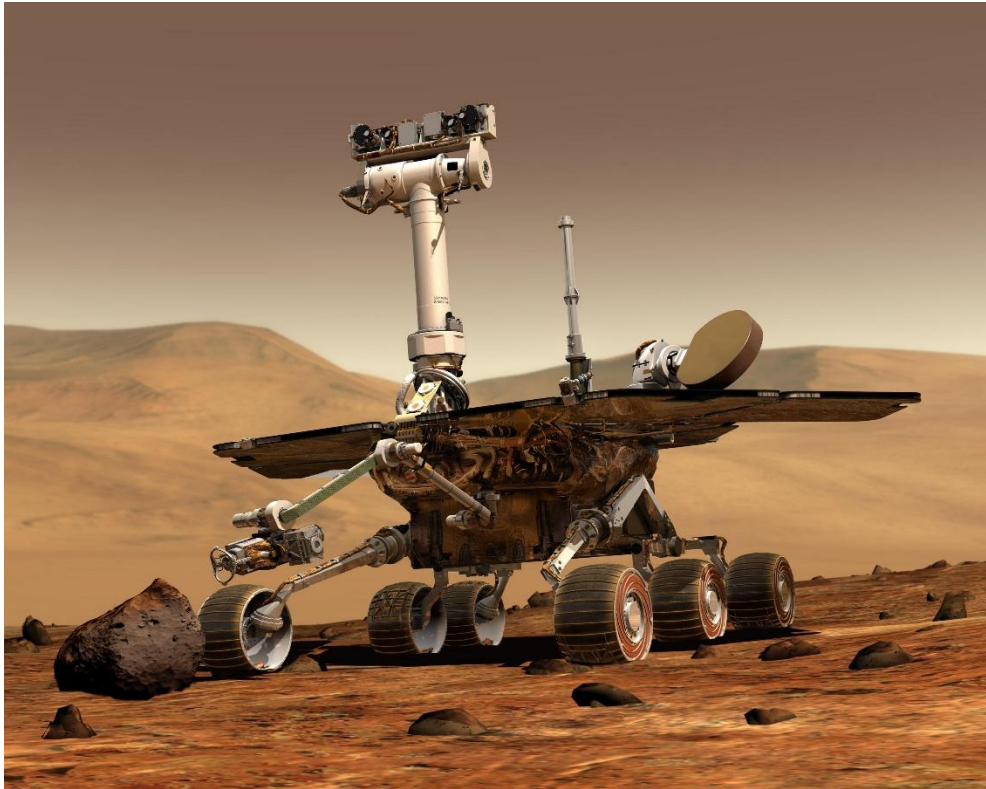


SoilSpirit

Autonomous No-dig Farming Rover



Appendix A

Creator:

Damon Rocha, Robotics and Embedded Systems

Last Updated: 3/15/2025

Technical Field

This project focuses on the following technical fields: robotics, embedded systems, autonomous systems, agriculture, and computer vision.

Background Information

No dig farming is a method of growing food where the soil is undisturbed. Instead of digging into the ground, compost is spread evenly on top of the soil and the seeds are directly put into this layer of material. It leads to healthy plants and soil, reduces weeds, and allows for better water conditions for the plants (Dowding, 2024). SoilSpirit plans to reduce the physical strain on small farms and gardeners through the automation of compost and seed spreading. While also having a small enough footprint that it does not compact the soil or cause unwanted tilling/digging of the compost. It takes inspiration from simple autonomous vehicles for obstacle avoidance and the Mars rovers (Spirit, Curiosity, Perseverance) for its rocker-bogie suspension system.

Prior Art (legal term)

The design of the product was heavily influenced by the Mars rovers and the 3d models of the initial prototype come from the Instructables tutorial in **AppendixC**. The suspension system of this rover comes from a direct iteration of the Rocker Bogie Mechanism popularized by the Mars rovers Curiosity and Perseverance. As for similar products, the closest to my idea is an autonomous spreader built by Raven Industries. It is an industrial vehicle that is autonomous. It is great for automation for large scale farms but would not benefit gardeners, homesteaders, and small-scale farms wanting to use no dig gardening. Another similar product that is of the same form factor/user is FarmBot. A CNC based garden bed that automates the process of growing food entirely. This company will be a direct competitor with the main difference being our modes of distribution. Their machine is directly attached to a garden bed whereas mine can drive around and be used for different types of gardens, farms, and homesteads.

Project Description

The aim of the project is to develop an autonomous rover capable of spreading fertilizer and seeds over a garden while avoiding obstacles. It should be able to:

- Drive on uneven surfaces
- Pull ~1lb of fertilizer
- ~~Operate on 6V~~ Changed 2/26/2025
- Operate on 12 V
- Evenly spread materials

- Autonomously stop rover when detecting obstacles using ultrasonic sensor **Changed 3/15/2025**
- Try to maneuver around obstacles using TOF sensor **ADDED 3/15/2025**
- Move ~ 2mph while not hauling spreader

It will consist of the following components:

1. Chassis to house electronics
2. 3 L298N motor driver **Changed to 3 L298N(using 2 added complications to the circuit and power management)**
3. 4 SG90 Servo Motors **Might be moving to 2 servos but depending on final wheel design**
4. 6 DC Motors
5. ESP32 ESP-S Dev Kit **Changed from Teensy4.1 Microcontroller on 2/22/25**
- ~~6. ESP32 XIAO S3 camera Removed due to pivot on 3/15/2025~~
7. 3.7V Lip rechargeable battery
8. Power bus with fuses **ADDED 3/15/2025**
9. VL53L5CX TOF sensor **ADDED 3/15/2025**
10. Ultrasonic sensor **ADDED 3/10/2025**
- ~~11. Object detection model~~
- ~~12. LLM for object detection~~
13. Hopper trailer with spring-based trap door to spread fertilizer/seeds

To demonstrate the system's capabilities, I plan on first running this on a mock test area in my garage that will consist of fake grass and various obstacles. Then I want to move on and try it with freshly cut grass in my backyard and get it to spread fertilizer until I have issued a stop command.

Innovation Claim

SoilSpirit will be the first adaptation of Mars rover rocker-bogie suspension for agriculture and will make automation accessible to more people. On top of that it promotes a sustainable agricultural practice in making no-dig farming easier. ~~Lastly, computer vision in a non-experimental realm will lead to this device being more robust and intelligent than most small-scale autonomous systems.~~ Removed 3/15/2025 due to pivot away for computer vision

Usage Scenario

The primary demographic of this product encompasses gardeners, homesteaders, and small farm owners who are interested in no-dig farming or already practice it. However, this system could easily be extended including applications across industry. In aerospace, this may very well become integrated with future iterations of rovers and space vehicles. Technology inspired by SoilSpirit could possibly be used for farming on other planets, if we ever get there, or as small-

scale versions of the Mars rovers that are more accessible for smaller companies. SoilSpirit could also extend into disaster recovery and reconnaissance. A scaled-up version of this rover could be used for search and rescue, area mapping, and debris clearing. Additionally, this project would be valued by the education and start-up industry due to it being a starting point for various projects.

Evaluation Criteria

The following questions will identify the successful completion of the project:

1. Core System Functionality:
 - a. Can the rover smoothly drive without user intervention? (DONE 2/26)
 - b. Can the rover move at least 2MPH while not hauling the hopper? (DONE 2/26)
 - c. Does the hopper start/stop spreading based on movement of the rover? (DONE 3/15)
 - d. Does the rover successfully detect obstacles and either avoid or stop before hitting them? (Stopping finished 3/15 avoidance dependent on TOF sensor integration)
 - e. Can the rover drive on uneven surfaces? (DONE 3/3)
2. Motor control and turning:
 - a. Can the motors drive the rover forward and backwards? (DONE 2/28)
 - b. Can the servos cause the rover to turn left/right? (DONE 2/26)
 - c. Is there enough charge to run the motors and the servos to allow for spreading to finish? (DONE 3/10)
3. Object Detection:
 - ~~a. Does the camera and model detect objects in front of the rover? (DONE 2/25)~~
 - ~~b. Does the camera and model detect objects to the left/right peripherals of the rover? (DONE 2/25) Removed due to pivot on 3/15/2025~~
 - c. Do the emergency stop ultrasonic sensors detect obstacles? (DONE 3/14)
 - d. Does a detection lead to the rover stopping? (DONE 2/28)
 - e. Does TOF sensor detect objects in different regions (Added 3/15)
 - f. Can the rover go around the object?
- ~~4. Documentation:~~
 - ~~a. Is the system and setup documented for ease of use and quickly getting started?~~
 - ~~b. Are limitations correctly represented by the documents?~~
 - ~~c. Is there a troubleshooting guide? (Removed 3/15 due to not being necessary for innovation)~~

Objectives and Tasks Associated with the Project

High-level Objectives

1. Development of Motor control and rocker-bogie suspension system
 - a. Create 2 rocker-bogie suspension systems **(DONE 2/22)**
 - b. Connect to microcontroller to test control **(DONE 2/24)**
 - c. Test DC motors, servos, and full suspensions systems individually then as a system **(DONE 2/22-2/24)**
 - d. Setup interface to drive motors based on shared commands ~~from ESP32 S3 Camera~~ **(DONE 2/28)**
2. Obstacle detection model created and integrated with ESP32 S3 Camera
 - ~~a. Develop object detection with ESP32 S3 Camera (DONE 2/22-2/26)~~
 - ~~b. Send detection results from XIAOS3 Cam to ESP32 ESP-S microcontroller over https—Changed from Teensy4.1 Microcontroller on 2/22/25, Changed from USART to wireless due to camera already calling LLM API(2/26/25) Removed 3/15 due to pivot~~
 - c. Test Ultrasonic sensor object detection and determine correct ranges **(DONE 3/10)**
 - d. Test detection of different objects, conditions, and distances **(DONE 3/10)**
 - e. Test TOF object region detection and obstacle cache **Added 3/15**
3. Rover control and drive system
 - a. Integrate object detection into motor control system
 - i. ON/OFF/start/stop/left/right commands from motor control system interface**(DONE 2/28)**
 - b. Setup start/stop/left/right/backwards commands user can send to rover from command system **(DONE 3/10)**
 - ~~c. Setup start/stop commands from ESP32 camera that a user can send to start autonomous control(DONE 2/28) Removed 3/15 due to pivot~~
 - d. Setup ultrasonic sensors as an emergency stop **(DONE 3/10)**
4. Sensor fusion
 - a. Create sensor fusion between ~~ESP32 S3 Camera~~ TOF output and ultrasonic sensors.
5. Spread of material
 - a. 3D Model Hopper Trailer **(DONE 2/28)**
 - b. Slice and print hopper trailer parts **(DONE 2/28)**
 - c. Setup trap door for material spreading **(DONE 3/3)**
 - d. Connect to **rover (DONE 3/3)**
6. Final rover design **(ADDED 3/15 to account for final items before SIP309)**

- a. Design modular rover chassis that fits all electronics and keeps them safe from the outside.
 - b. Design planetary gear system to turn the wheels from DC motor output
(ADDED due to issues with wheel design I got from the instructables website, gears are stripping and the system as a whole is too fragile for extended usage)
 - c. Integrate TOF sensor into obstacle detection for rover
7. Test full system integration

Detailed Task Breakdown

To start I plan on building out the motor control and object detection in tandem while I am waiting for a few final parts for the drive system. I plan on building up the object detection using an Ultrasonic sensor as the base which will be used to detect if the rover needs to stop and then a TOF sensor to determine whether or not the rover can try to navigate around the obstacle. ~~of the XIAOS3 Cam by connecting it to an LLM and feeding the image captured to the LLM by sending the base64 encoded version to an API. I plan on starting with Claude, and then moving over to a local open source LLM I want to setup on my local router network, if time permits.~~ (REMOVED 3/15 due to pivot) ESP32 cam using the FOMO(Faster Objects, More Objects) Object detection from Edge Impulse. ~~If I run into issue with the AI model or run into walls with performance I will revert to classic computer vision algorithms and do some feature extractions for less precise but still working object detection.~~ (Changed 2/26/2025) I want to first test these systems by themselves and integrate them together with the ESP32 ESP-S (Changed from Teensy4.1 Microcontroller on 2/22/25). Once I have this to a working state, I plan on moving to communication between the boards using ~~USART~~ HTTPS while modeling the hopper trailer. I am going to build a basic communication interface that consists of basic commands dependent on object detection results and user commands from command app ~~where they are in the image~~ (Changed 3/15). I will do various system tests and integration tests to ensure everything is working individually and together. Afterwards, I plan on finishing up the design/build of the rover chassis and a better drive system for the wheels. Once I finish these parts I will test out the rover on its own and attach the 3 tow hitches to connect to the hopper trailer. Then finally finish building the hopper and test out full system integration.

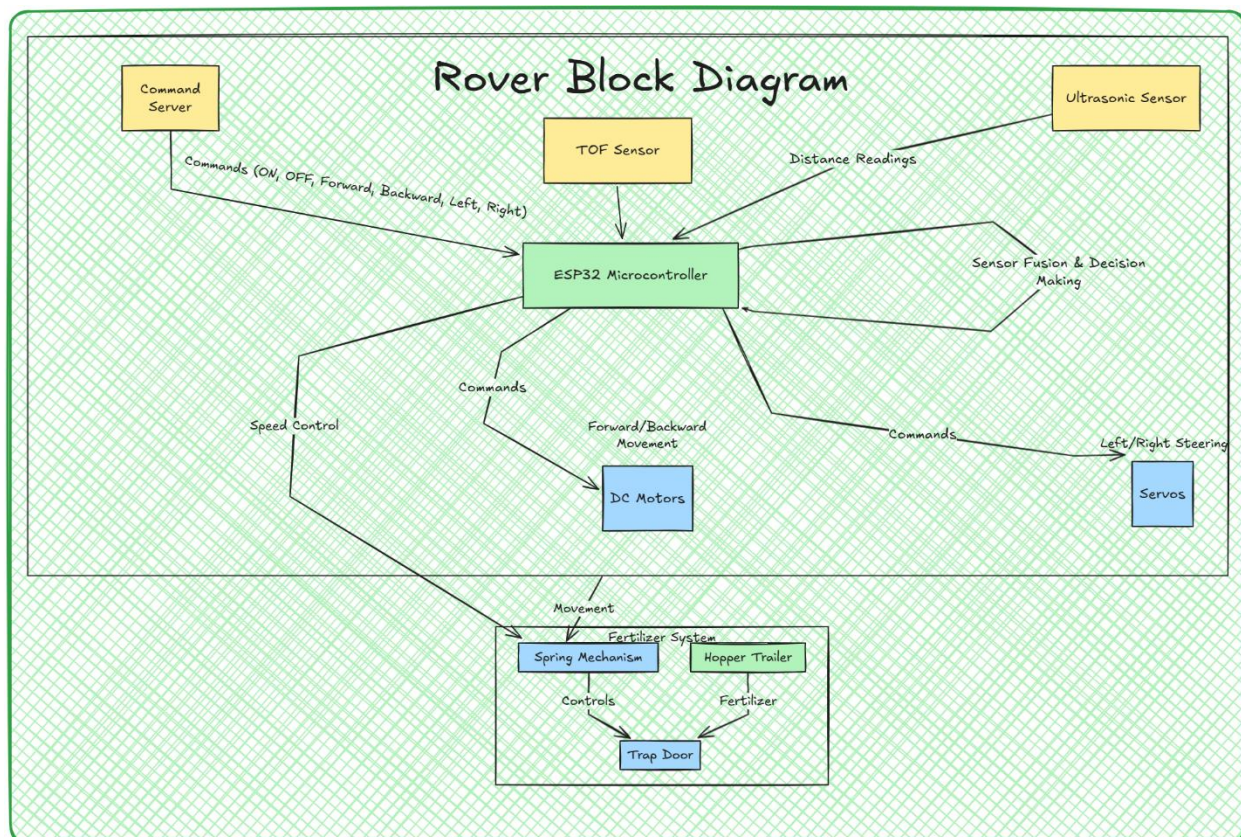
Description of Design Prototype

Overview

The main devices of this project will consist of the ESP32 ESP-S (Changed from Teensy4.1 Microcontroller on 2/22/25) microcontroller ~~connected through USART~~ (Changed on 2/26/2025) which will pull commands from a command server that sits on a ZimaBoard NAS which will

store commands users provide such as start, stop, and directional commands to the rover. These commands will come from an app I plan on developing which for now will just be an inline HTML application so that I can easily test it out on my phone. (ADDED 3/15) ~~the ESP32-S3 Camera module based on image detection from an LLM(REMOVED 3/15)~~. The ESP32 ESP-S microcontroller will use 3 L298N Motor Drivers to power the DC motors that will drive the wheels. For turning 4 servo motors are mounted on the front and back wheels which will be connected to the ESP32 Dev Board powered using pulse width modulation on Left/Right commands. The DC motors will be powered ON/OFF/Forward/Backwards depending on commands from the server ~~ESP32-Camera~~. As a backup object detection system there will also be an ultrasonic sensor used for emergency object detection in case that the User tries to drive the rover into something or there is a command bug. (ADDED 3/15) As stated earlier the mechanical design of the rovers rocker-bogie suspension system was predesigned but I only plan on using the suspension and will be using a custom chassis and slightly different electronics. I also will need to redesign the wheel drive system due to gear stripping and not working for long usage outdoors (ADDED 3/15).

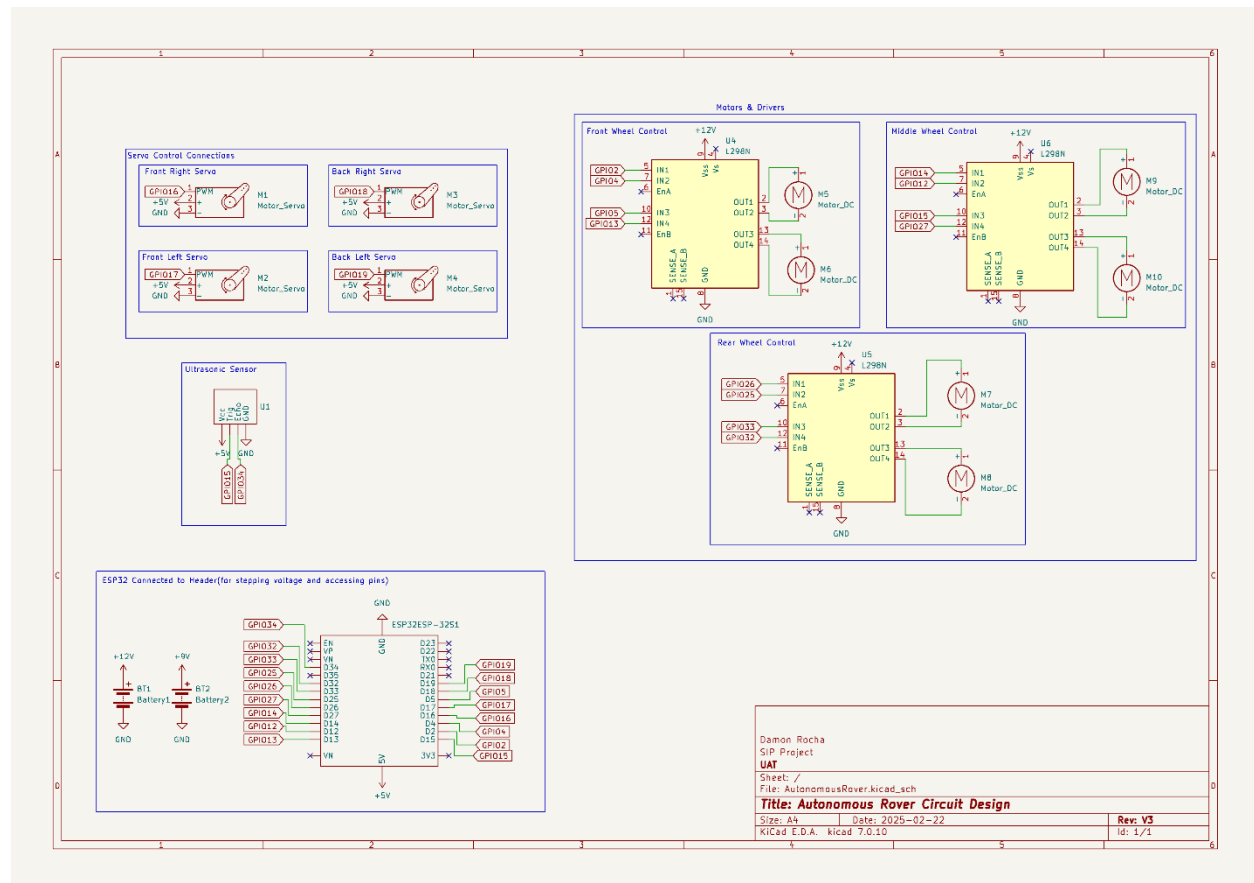
As for the Hopper trailer, this will be 3D printed and contain no electronics. It will consist of a hopper module that feeds the fertilizer/seed to a trap door which is opened/closed based on a spring which will be connected to the rover so that upon it moving the trap door will open/close.



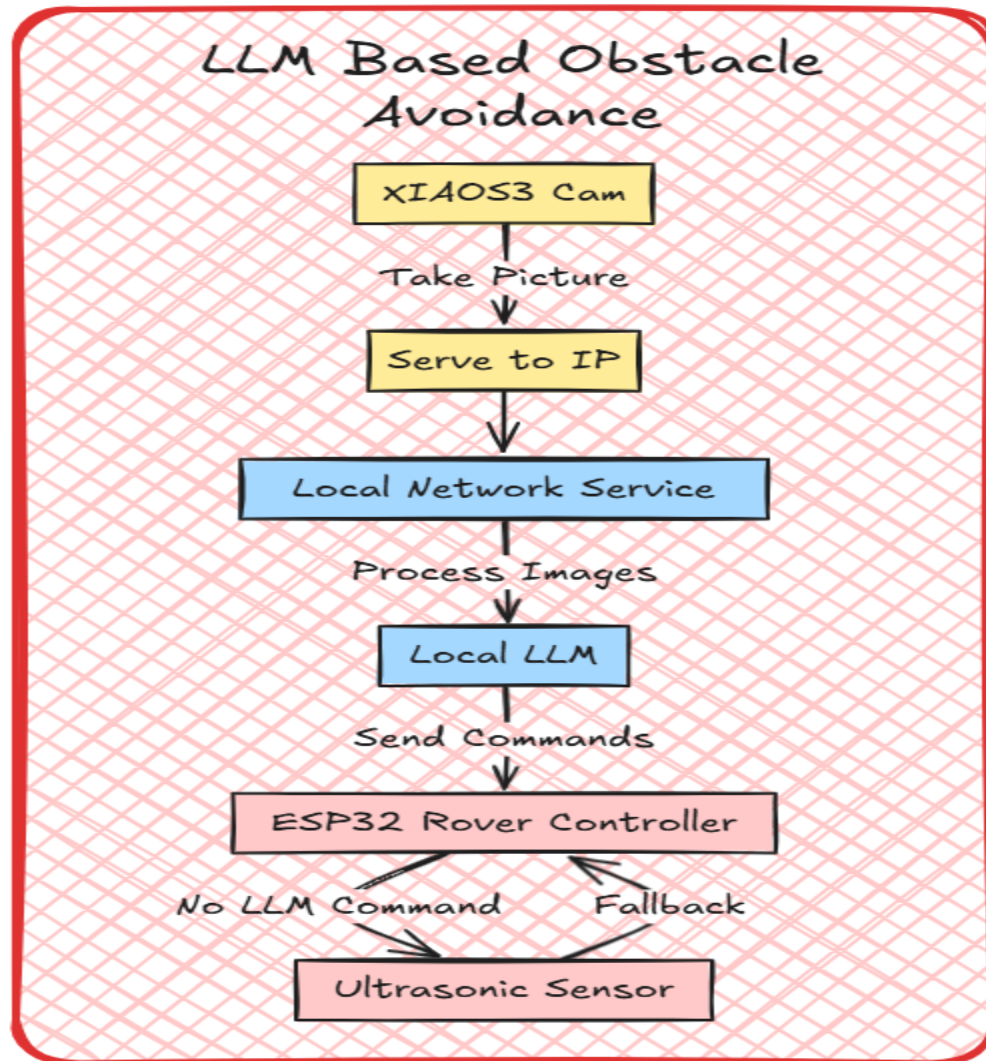
Electronics

I started the circuit design with the schematic below. However, I soon found that the 6V supply was not enough to get my rover moving on carpet let alone grass. I did not, however, have a 12V supply initially so I powered my initial prototype by wiring 2 6V packs in series and using a 9V battery to power the servos/microcontroller. As for the XIAO Camera I have a small rechargeable LiPo battery that I plan on plugging into some leads I soldered to the XIAO Camera. (Added 3/3/2025)

The command queue server is Zimaboard NAS that hosts the server using a docker image. This allows the rover and camera to easily communicate through the NAS which was much more stable than the communications between the devices themselves. (Added 3/3/2025)



Appendix C (ADDED 2/22/25)(UPDATED 3/15)



~~Appendix D (Added on 2/26/25)~~ (REMOVED 3/15)

Software

As for the software side of things I will use the following repo to house my code, designs, and documents as I build them up: <https://github.com/dmarcr1997/SoilSpirit>. I expect to separate my code into 2/3 main directories depending on the LLM those are:

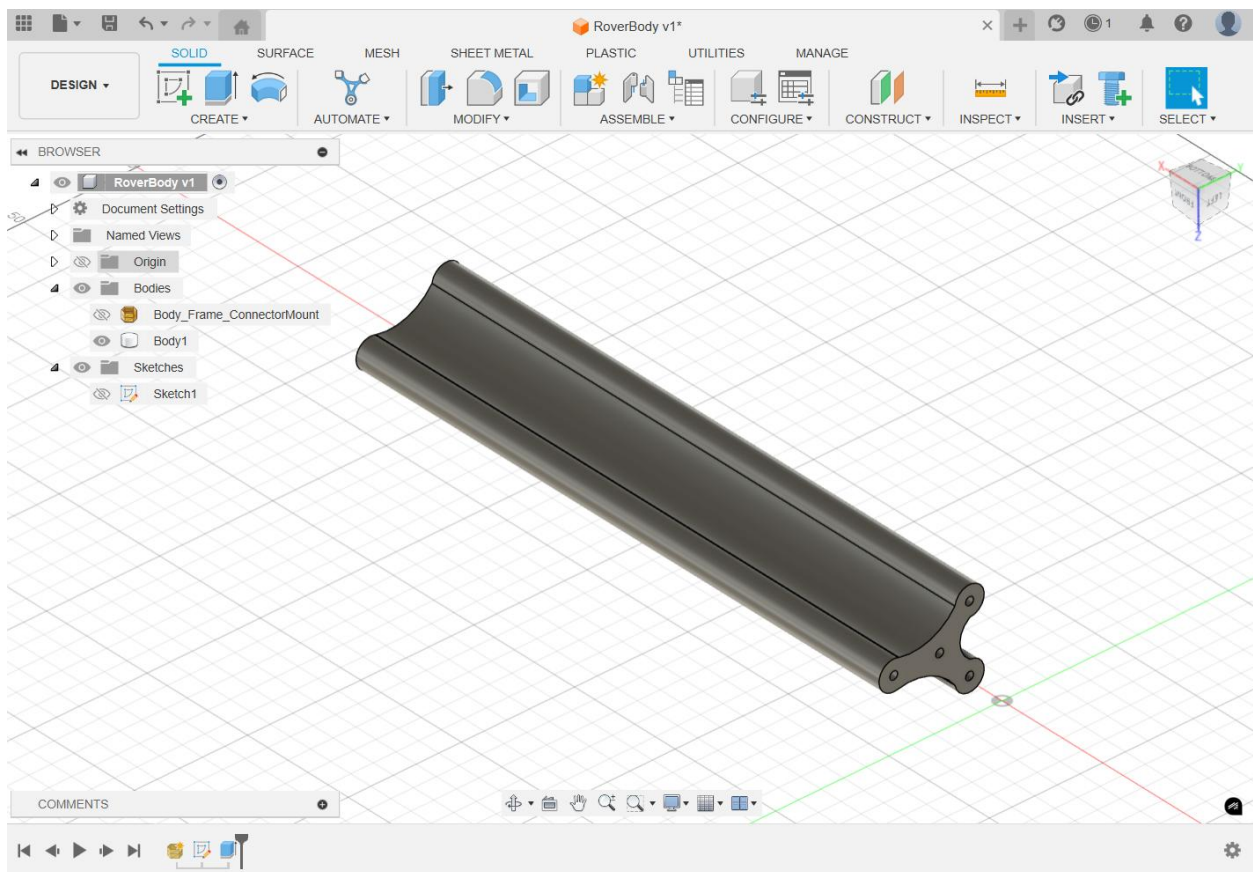
- ~~1. XIAO Camera Object Detection Code~~ (REMOVED 3/15 due to pivot)
2. Command Server (ADDED 3/15)
3. Controller APP (ADDED 3/15)
4. ESP32 32S Motor Driver

And have sub directories consisting of test programs and the overall final logic for the camera and rover. (Added 2/26/2025).

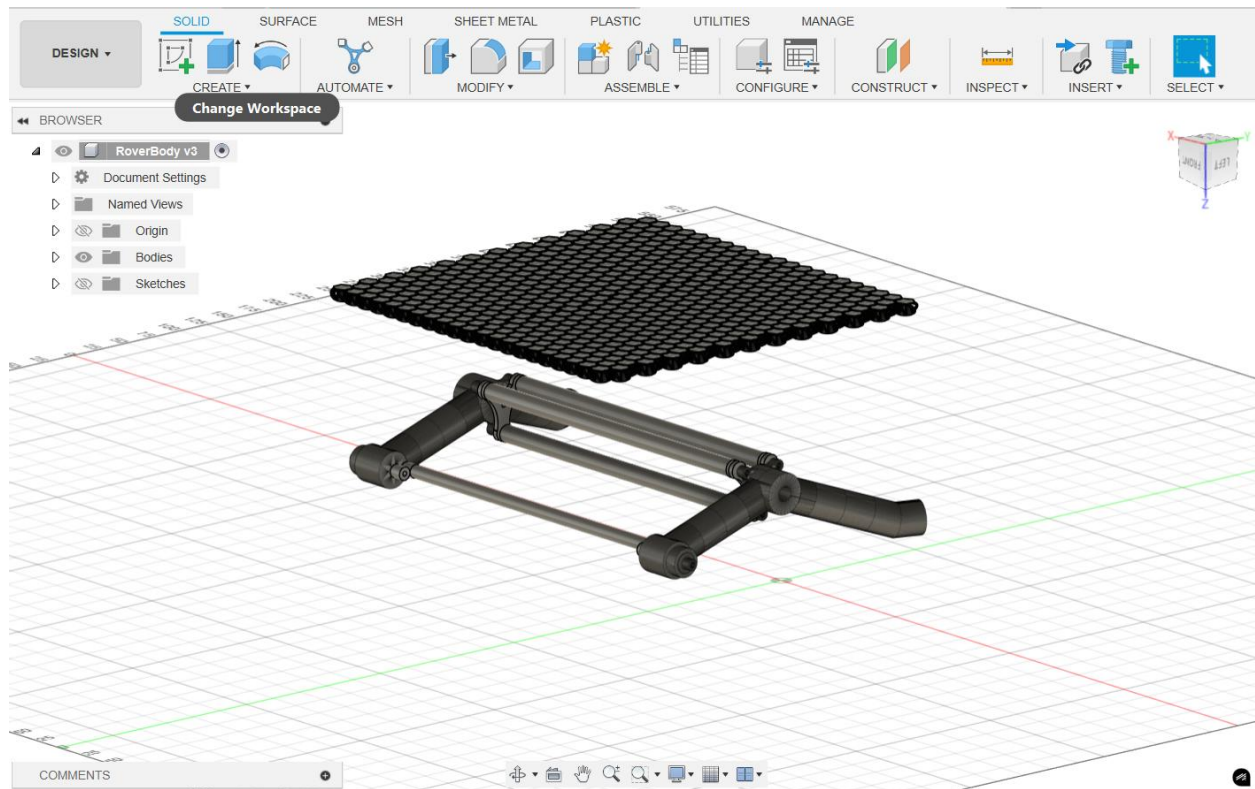
Body

I have also made some design modifications to the original rover design I got from the instructables in the Appendix C. The frame initially did not work well with a soft rover body. I tested this with cardboard and found that the suspension bowed and would not allow the wheels to turn.

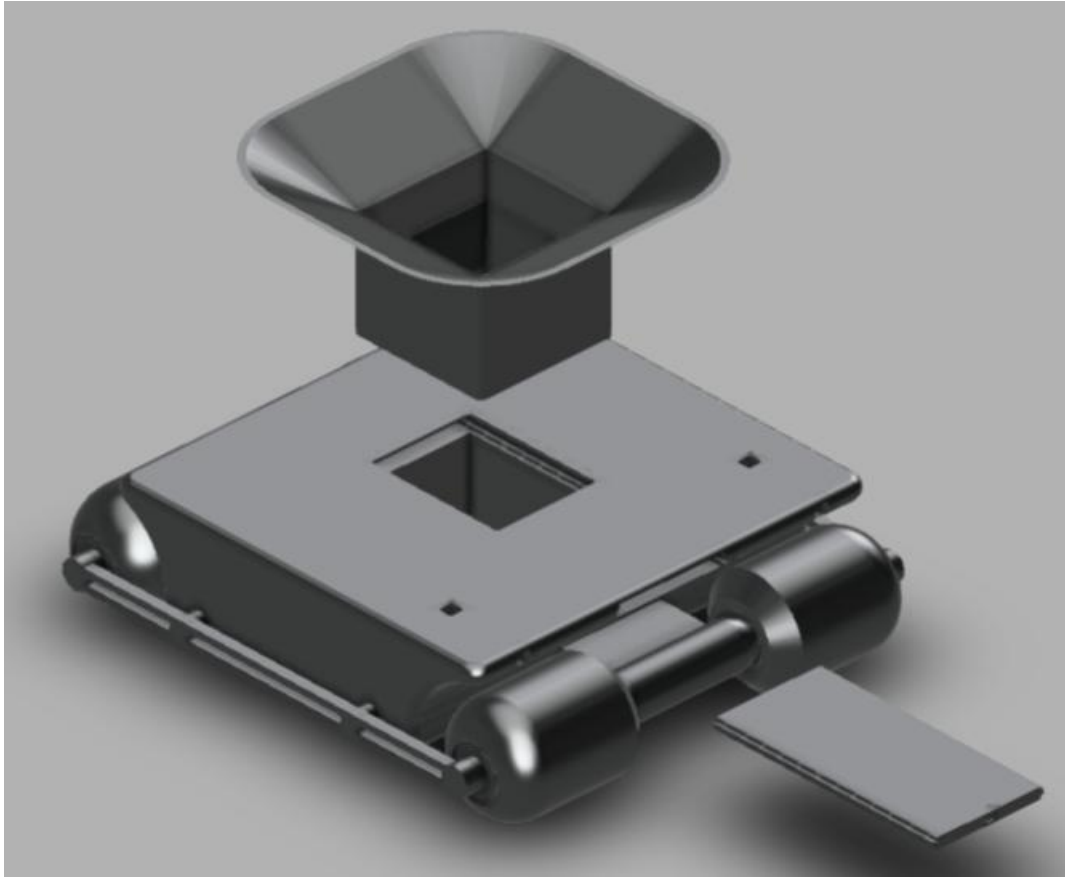
I started with a single support that went from body connector to body connector on the interior and then had a whole bunch of other ideas. I then wanted to try and build a smaller frame that is more ergonomic and easier to replace, but that is a major stretch goal abit outside of the scope of the innovation on this project. If I get to this point, I plan on using some of the 3d printed NASA hexagon fabric to cover the chassis and allow it to have a soft body which is easy to replace while still being really strong. (ADDED 2/26/2025)



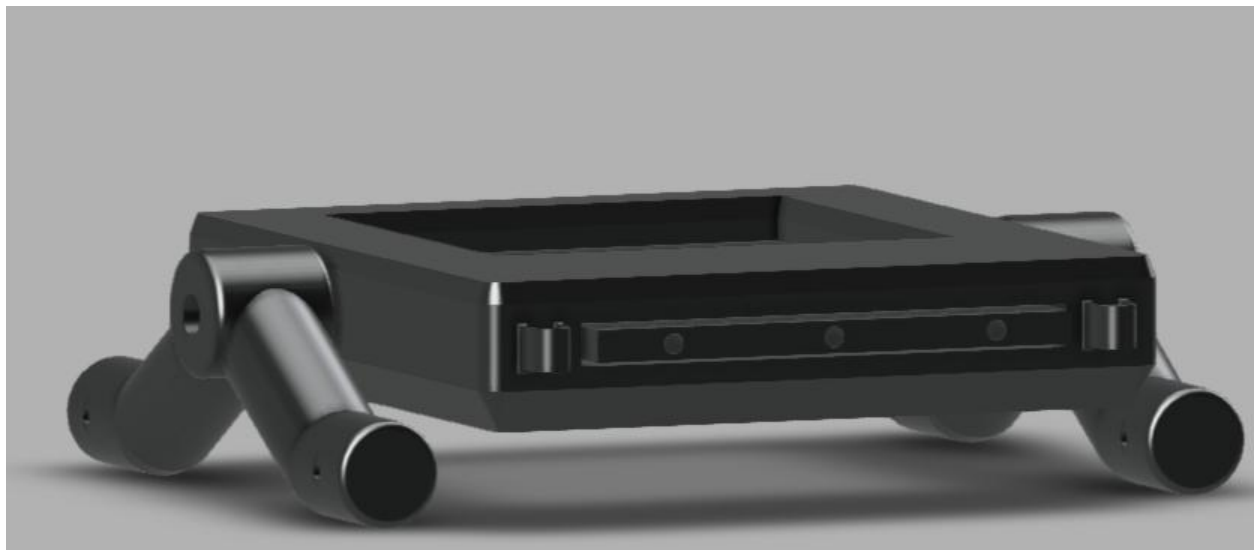
Appendix D (ADDED 2/26/2025)



Appendix E (ADDED 2/26/2025)



Appendix F (ADDED 3/16/2025)



Appendix G (ADDED 3/16/2025)

On top of the body redesign, I found that the wheel drive system which consists of 2 gears strips rather easily so I had to redesign the wheel system. I plan on adding more info here as I get that done, and am thinking of looking into a planetary gearbox system instead. (ADDED 3/15).

Evaluation Plan

This project will go through testing and evaluation across many different levels. Each system will be broken into modular parts and tested independently (unit tests) and in junction with other systems (integration tests). Each section of the evaluation criteria represents one system and will be tested as such. So, for example, the motor control of each wheel module will be tested independently and then together, then as a whole for the rover, then also integrated as part of the full autonomous system. On top of this the requirements I stated at the beginning of this brief either need to be met or show areas that need improvement. For clarity, here those are again:

- Drive on uneven surfaces
- Pull ~1lb of fertilizer
- Operate on 12V
- ~~Operate on 6V (Changed 2/26/2025)~~
- Evenly spread materials
- Autonomously avoid obstacles using camera and ultrasonic sensor
- Move ~ 2mph while not hauling spreader

Project Completion Assessment

Note: This section must be completed prior to SIP403/409.

Provide an in-depth description of the completion assessment of your project. Describe how well the completed components function and highlight the innovative facets of your design. This is sometimes known as a “Post-Mortem” or “Lessons-Learned Report”. A good approach for this section is to answer the following 4 questions: “What went right? What went wrong? What was learned throughout the process? What would be done differently if you had to do it again?”

Appendices

Appendix A: Picture of Spirit Mars Rover – SpiritRover.jpeg

Appendix B: Rover Block diagram – RoverBlockDiagram.png

Appendix C: Circuit Design Schematic – SIPCircuitDesign.png

~~Appendix D: LLM Object Detection Block diagram – LLMOBJBd.png (REMOVED 3/15 due to pivot)~~

Appendix D: Rover Body Support CAD model – RoverBodySupportCAD.png

Appendix E: RoverBodyV3 CAD model – RoverBodyV3CAD.png

Appendix F: Hopper Trailer V1 CAD model – TrailerModelV1.png

Appendix G: Hopper Trailer Final CAD model – HopperTrailerSide.png

Appendix C: References

Dowding, C. (n.d.). *Beginner's guide to no dig gardening*. Charles Dowding. Retrieved February 18, 2025, from <https://www.charlesdowding.co.uk/resources/beginners-guide>

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Printables. (n.d.). XIAO Sense Case - Seeed Studio XIAO ESP32S3 Sense Case. Retrieved March 4, 2025, from <https://www.printables.com/model/681623-xiao-sense-case-seeed-studio-xiao-esp32s3-sense-ca/files>

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